

F1 / A45 — "HelixCode session crashes the whole terminal" — RESOLUTION

Revision: 1 **Last modified:** 2026-06-13T18:30:00Z **Status:** RESOLVED (operator-confirmed "works now", 2026-06-13). Closed to Fixed.md A45.

Root cause (REPRODUCED, not guessed — §11.4.6/§11.4.123)

The installed/generated wrapper scripts/tmx had:

```
TMUX_BIN="/Users/milosvasic/Projects/tmux/tmux/build-darwin/bin/tmux"
```

— a path from a PRIOR checkout location that **does not exist** on the live host (the live checkout is /Volumes/T7/Projects/tmux).

The operator's shell-init runs the operator path:

```
exec sh -c 'tmx attach -t HelixCode 2>/dev/null || exec tmx new -s HelixCode'
```

tmx new -s HelixCode reaches exec "\$TMUX_BIN" ... (scripts/tmx.template lines 396 / 430) on the MISSING binary → exec fails with exit 127 → the operator's **login shell** (which had exec'd into the wrapper chain) is replaced by the failed exec and **dies** → the terminal window closes = "crashes the whole terminal." It is the shell dying, so it reproduces identically on every emulator (iTerm2 / Terminal.app / Linux / WezTerm), matching the operator's all-emulators report.

Captured proof (real PTY, this host)

```
$ # current wrapper after v1.0.22 setup.sh
TMUX_BIN="/Volumes/T7/Projects/tmux/tmux/build-darwin/bin/tmux"
exists? YES
```

```
$ # bad-TMUX_BIN wrapper driven through the EXACT operator path
over a PTY:
PTY EOF – controlling shell DIED (terminal would close = the
crash)
child exit status: 127
captured: ".../tmx: line 396: /Users/milosvasic/Projects/tmux/
tmux/build-darwin/bin/tmux: No such file or directory"
```

Why the earlier headless forensics did not catch it

A *fresh* `tmx new -s HelixCode` on THIS checkout (correct `TMUX_BIN`) created + attached cleanly. The 5 `tmux/config-layer` crash vectors (`passthrough`, `extended-keys`, `attach-reload`, `rename-format`, `stale socket`) were correctly **DISPROVEN** (see `forensic.md`). The defect lived in the **wrapper-generation layer** (`stale TMUX_BIN`), present in the operator's environment but not on the fresh checkout the headless tests used.

Fix

`scripts/setup.sh` regenerates `scripts/tmx` from `scripts/tmx.template`, substituting the correct `__TMUX_BIN__` for the live checkout. Running `v1.0.22` `setup.sh` rewrote the operator's wrapper with the valid path → `exec` succeeds → no crash. **Operator-confirmed: "works now."**

Regression guard (4-layer, §103 / §11.4.135) — so it can NEVER recur silently

Layer	Artifact	What it catches
1 source gate	<code>verify.sh</code> <code>CM-TMX-WRAPPER-TMUXBIN-VALID</code>	a present <code>scripts/tmx</code> whose <code>TMUX_BIN</code> path is missing/non-exec → <code>verify/setup</code> REFUSE to bless it
3 runtime	<code>scripts/tests/60_wrapper_tmux_bin_valid.sh</code>	T1 static valid; T2 RED reproduces the missing-binary <code>exec</code> failure via the operator path; T3 GREEN valid wrapper creates the session. <code>PASS=3/0/0 ×3</code>
4 paired mutation	<code>M-WRAPPER-TMUXBIN</code> (meta)	rewrite <code>scripts/tmx</code> <code>TMUX_BIN</code> → missing → test 60 FAILs (CAUGHT). Meta sweep 52/0/8 GREEN

Sources verified 2026-06-13

- Reproduced on host (macOS 15.5 / arm64); evidence captured this session.